Eric Atanga Ayamga

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Research Interests

Applied Microeconomics, Applied Econometrics, Industrial Organization, Health Policy, Energy Transition

Education

Texas Tech University, Lubbock, TX — Expected June 2027
Ph.D. in Economics
Thesis:
Committee: Michael Noel (Chair), Victoria Ziqi Hang (co-chair), Misak Avestiyan

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana — 2022
MPhil. in Economics

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana — 2019
B.A. in Economics

References

Michael D Noel
Professor of Economics
Department of Economics, Texas Tech University
Email: Michael.Noel@ttu.edu

Misak Avestiyan
Associate Professor and Associate Chair of Economics
Department of Economics, Texas Tech University
Email: Misak.Avestiyan@ttu.edu

Victoria Hang
Assistant Professor of Economics; Associate Graduate Program Director
Department of Economics, Texas Tech University
Email: Victoria.Hang@ttu.edu

Working Papers

Do Front-of-Package Sugar Labels Shift Consumer Demand away from Diabetes-Risk Foods? (Job Market Paper), submitted

This paper evaluates the impact of a U.S. front-of-package (FOP) sugar labeling policy on consumer demand for ready-to-eat (RTE) cereals. Using NielsenIQ retail scanner data and a difference-in-differences design, we find no evidence of product reformulation or significant demand shifts around the 5- and 10-gram added sugar thresholds following the introduction of the table-format FOP label. Although we detect a modest substitution pattern beginning around 7.5 grams of added sugar, the magnitude is economically small and unlikely to meaningfully affect diet-related health outcomes. To rationalize these limited effects, we develop and estimate a structural Berry–Levinsohn–Pakes (BLP) demand model augmented with a rational inattention framework in which information processing costs enter utility directly. We allow the cost of processing nutritional information to vary by label format, with the highest cost associated with table-format labels, followed by letter-grade labels, color-grade labels, and the lowest cost for warning labels. Structural estimates indicate that high information-processing costs substantially attenuate the effectiveness of table-format labels. Counterfactual simulations show that simplified formats particularly warning labels significantly reduce demand for high-sugar cereals and generate welfare gains. Overall, our findings suggest that reducing cognitive costs through simplified FOP designs can meaningfully improve consumer welfare and shift consumption away from unhealthy products. Paper link

The Cost of Inaccessibility: Retail Discrimination and Mobility-Constrained Consumers (With Li Zeng and Victoria Hang)

This study investigates whether individuals with mobility-related health conditions face systematic disadvantages in retail pricing, both in physical stores and on online platforms. While price discrimination is a common feature of modern markets, limited attention has been paid to how physical mobility constraints may affect consumers' ability to access lower prices or respond to promotions. Using matched observational data from the NielsenIQ and the Open E-Commerce dataset, we compare purchasing behavior and price outcomes between consumers with and without mobility-related health conditions. We find that mobility-constrained individuals pay modestly higher prices in physical stores and face even larger price disparities in online purchases, particularly among wheelchair users. Our findings highlight mobility as an underexamined axis of vulnerability in consumer markets. The results have implications for the design of retail pricing systems and digital platforms, as well as for policy efforts aimed at improving equity in access to essential goods.

Work in Progress

Temperature Shocks and Diabetes Risk Exposure

This study examines how household consumption of sugar-sweetened beverages in the US is impacted by rising temperatures. We calculate the causal impact of weekly temperature variations on the demand for sweet products using Nielsen retail scanner data from convenience stores and county-level temperature data. We find a strong, positive correlation between temperature and the consumption of sugary drinks using a fixed-effects model that accounts for endogeneity through instrumental variables, such as offshore temperature and geographic features. Increased demand is linked to warmer weather, which may be caused by situational and behavioral factors like mood swings, impulsive purchases, and increased outdoor activity. These findings highlight how behavioral consumption responses must be taken into account when evaluating the wider economic and health effects of climate change.

Research Publications

- Decomposed Oil-Driven Inflation and Asymmetric Shocks. Published in **Studies in Nonlinear Dynamics & Econometrics** (with Joseph Agyapong)
- Forecasting the Future: Applying Bayesian Model Averaging for Exchange Rates Drivers in Ghana. Published in **Applied Economics** (with J. Agyapong and S.I. Anyars)

- The Macroeconomic Impact of Energy Transition Metal Prices: Nonlinear Evidence from IEA Economies. Available in **SSRN** (with J. Agyapong and Lordia Yalley)
- The Environmental Impact of International Trade in Sub-Saharan Africa: Exploring the Role of Policy and Institutions for Environmental Sustainability. Published in **Research in Globalization** (with D. Mpuure, E. Doudu, and AB. Abille)
- Pricing of Climate Risks in Capital Markets of South Africa. Published in **Journal of Developing Areas** (with E. Amporfu and D. Sakyi)

Referee Services

Referee: *International Economics and Economic Policy* (5 papers)
Referee: *South African Journal of Economics* (1 paper)

Teaching Experience

Instructor

- TTU - Eco 2301 Principles of Microeconomics — Fall 2025, Spring 2026

Teaching Assistant

- TTU - Eco 6312 Microeconomics II (Game Theory) — Spring 2024
- TTU - Eco 5311 Macroeconomic Theory and Policy — Fall 2023
- KNUST- Econ Microeconomic Theory I, — Semester I, 2022
- KNUST Mathematical Economics, — Semester I, 2022
- KNUST Econometric Theory I, — Semester II, 2022
- TTU Monetary Economics, TTU — Spring 2025
- TTU - Eco 3305 Game Theory, TTU — Spring 2024
- TTU Principles of Money, Banking, and Credit, TTU — Fall 2023
- TTU Intermediate Macroeconomics, TTU — Spring 2023
- TTU Principles of Microeconomics, TTU — Fall 2022
- KNUST - Econ 479/480 Advanced Econometrics I & II — Fall 2019, Spring 2020
- KNUST - Econ 463/464 Environmental and Resource Economics I — Fall 2019, Spring 2020

Professional Experience

Graduate Part-Time Instructor, Texas Tech University, Texas

- Teach undergraduate courses in Economics
- Assign and grade homework and exams

Teaching Assistant, Texas Tech University, Texas

- Assist Professors teach, assign assignments, and grade assignments
- Offer tutoring services to students

Research Assistant, Texas Tech University, TX

- Worked on NBER-supported research for Faculty
- Conducted econometric analysis using Matlab and Stata

Teaching Assistant, KNUST, Kumasi, Ghana

- Assist teach graduate students Microeconomic Theory I and Econometric Theory I

Teaching Assistant, KNUST Kumasi, Ghana

- Tutor undergraduate students, assign assignments, and grade assignments and exams

Research Assistant, Technology Consultancy Centre - KNUST, Kumasi, Ghana

- Manage survey and experimental data using Python

Banking Intern, Access Bank Ghana Limited

- Perform anti-money laundering (AML) and know your customer (KYC) risk analysis
- Analyse product reports and develop analytical tools

Honors and Awards

- Graduate Part-Time Instructor, Texas Tech University — Fall 2025
- Dr Rashid Al-Hmoud Fellowship, Texas Tech University — 2022, 2023, 2024, 2025
- Graduate Students Research Support Fund, Texas Tech University — 2025
- Faculty Dean Award, Kwame Nkrumah University of Science and Technology — 2019
- Best Economics Student Award, Kwame Nkrumah University of Science and Technology — 2019
- Best Economics Student Award, Kwame Nkrumah University of Science and Technology — 2018

Conference Presentations

Midwest Economic Association Conference	2025
ASSA/AEA Annual Meeting	2024, 2025
International Studies Association Annual Meeting	2025
Arts and Humanities Annual Conference	2023, 2024, 2025

Skills

Programming and Software: Stata, Python, R, MATLAB, Excel, EViews, HTML, LaTeX